Start-of-lesson routines - Mathematics 2G

school year 2026-2027 - Teacher edition

> One sheet per 2G session (88 sheets, from 8 October 2026 to 1st July 2027), aligned with the calendar; questions 4 and 5 take into account the diagnostic test of 29 September 2026 (see the guide and Appendix A).

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Vectors in the plane

Sheet 1 Thu. 8 Oct. 2026 Distance between two points and the midpoint of a segment

1. Warm-up · Flash [CAL] Calculate $(-3)^2$, $(5-(-2))^2$ & -3^2.

Ans. $9\,;\,49\,;\,-9$.

Prerequisite for calculating a distance: the square of a signed number.

2. Core · Method [GEO] $A(1;2)$ and $B(4;6)$ in an orthonormal coordinate system. Give the coordinates of $\overrightarrow{AB}$ then the distance AB.

Ans. $\overrightarrow{AB}\begin{pmatrix}3\\4\end{pmatrix}$; $AB=\sqrt{3^2+4^2}=5$.

Pitfall: $x_A - x_B$ instead of $x_B - x_A$; forgetting the square root.

3. Core + · MCQ [REP] The midpoint of $[AB]$ with $A(-2;1)$ and $B(4;-3)$ has coordinates:

A. $(1;-1)$ **B.** $(-1;1)$ **C.** $(2;-2)$ **D.** $(3;-2)$

Ans. A. $(1;-1)$

Distractors: half-difference $(3;-2)$; undivided sum $(2;-2)$; sign error $(-1;1)$.

4. Review · Flash [SUB] Calculate $3x^2-2x+1$ for $x=-2$.

Ans. 17.

5. Running thread · Flash [EQ] Solve in $\mathbb{R}$: $3x+5=17$.

Ans. $x=4$

Sheet 2 Mon. 12 Oct. 2026 Determinant of two vectors; collinearity criterion

1. Warm-up · Flash [CAL] Calculate $2\times(-3)-4\times 5$ and $(-1)\times 6-3\times(-2)$.

Ans. -26 ; 0.

Determinant routine: cross-multiplying signed numbers.

2. Core · Method [GEO] Calculate $\det(\vec{u},\vec{v})$ for $\vec{u}\begin{pmatrix}2\\-3\end{pmatrix}$ & $\vec{v}\begin{pmatrix}-4\\6\end{pmatrix}$. What can we conclude?

Ans. $2\times 6-(-3)\times(-4)=12-12=0$: $\vec{u}$ and $\vec{v}$ are collinear ($v=-2\vec{u}$).

3. Core + · True/false [GEO] True or false? Justify. The vectors $\vec{u}\begin{pmatrix}1\\2\end{pmatrix}$ & $\vec{v}\begin{pmatrix}3\\5\end{pmatrix}$ are collinear.

Ans. False. $\det(\vec{u},\vec{v})=1\times 5-2\times 3=-1\neq 0$.

4. Review · Construction steps [CON] Draw a circle with centre O and two perpendicular diameters $[AC]$ and $[BD]$. What kind of shape is $ABCD$? Justify.

Ans. A square: diagonals with the same midpoint O (parallelogram), of equal length (rectangle) and perpendicular (rhombus).

5. Running thread · Flash [ALG] Procedure: choose a number; multiply it by 4; add 5. a. What result for 3? b. What expression in terms of the starting number x ?

Ans. a. 17. b. $4x+5$.

Sheet 3 Tue. 13 Oct. 2026 Collinearity and parallelism: the determinant method

1. Warm-up · Flash [GEO] Calculate the coordinates of $\overrightarrow{AB}$ for $A(1;-2)$ and $B(4;3)$.

Ans. $\overrightarrow{AB}\begin{pmatrix}3\\5\end{pmatrix}$.

2. Core · Method [GEO] Are points $A(1;2)$, $B(3;3)$ and $C(7;5)$ collinear?

Ans. $\overrightarrow{AB}\begin{pmatrix}2\\1\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}6\\3\end{pmatrix}$: $\det\left(\overrightarrow{AB},\overrightarrow{AC}\right)=2\times 3-1\times 6=0$: collinear.

3. Core + · Method [EQ] For which value of k are vectors $\vec{u}\begin{pmatrix}k\\3\end{pmatrix}$ & $\vec{v}\begin{pmatrix}4\\6\end{pmatrix}$ collinear?

Ans. $\det(\vec{u},\vec{v})=6k-12=0$, therefore $k=2$.

The determinant leads to a first-degree equation.

4. Review · Flash [PRO] Calculate 20% of 350.

Ans. $0{,}20\times 350=70$.

5. Running thread · Translate [LIT] A rectangle has length $x+3$ and width x (in cm). Express its perimeter as a function of x and simplify it.

Ans. $P=2((x+3)+x)=4x+6$ (in cm).